

APPENDIX W

SOURCE CODE LISTING OF PERSONAL COMPUTER PROGRAM FOR 5 pH POINT TITRATION METHOD

This appendix contains the listing of the source code for the source code file: 5ppt.pas. This file is available on the floppy disk which is attached to the inside of the thesis cover. The program is coded using Turbo Pascal Ver 4.0. It allows the calculation of H_2CO_3^* alkalinity, SCFA (as A_T) and systematic pH measurement error from the data collected in the 5 pH point titration procedure. (For 5 pH point titration procedure, see Appendix V).

```
program atct;
```

``` (5 pH POINT TITRATION PROGRAM) ```

(This program calculates H₂CO₃-alkalinity and short-chain fatty acid concentrations in aqueous solutions containing the carbonate, SCFA, ammonium and phosphate subsystems. The total species concentrations of the ammonium and phosphate subsystems need to be entered into the computer program. The sample is titrated from its initial pH to four further pH values measured in the pH region: 6.7, 5.9, 5.2, 4.3. The program provides an estimate of a possible systematic pH measurement error in the event that the carbonate system dominates over the remaining weak acid/base subsystems.)

```
(PROGRAM START:)
```

```
uses crt;
```

```
type phstr= string[50];
```

```
var pH0, pH1, pH2, pH3, pH4, delph, pHcorr,
    A1, B1, A2, B2,
    Mct1, Mct2, Ct2, Ct1, Ctcomp, Mat1, Mat2, At1, At2, CtAtratio,
    dil, vx1, vx2, vx3, vx4, vxfi, vxs, ca, temp, ktemp, TDS,
    logf1, logf2, pK1, pK11, pK2, pK22, pKn, pKnn, Nt,
    pKp, pKpp, Pt, pKa, pKaa, At,
    MH2CO3alksam, H2CO3alksam, Mcarb01, Mcarb02,
    counter, ionstr, number, vsdil, vsundil :real;

    again, move,intro, x, y, a, b, c, d : char;
    labels: array[1..18] of phstr;
    value: array[1..18] of real;
    I,II,posy: Integer;
```

```
function log(x:real):real;
begin
    log:=ln(x)/2.302585093;
end;
```

```
function tento(y:real):real;
begin
    tento:=exp(y*ln(10));
end;
```

```
function mue (TDS,dil:real):real;
begin
    mue:= 0.000025* (TDS/dil-20);
end;
```

```
function logf (mue,ktemp:real):real;
begin
    logf:= - 1.825 * 1000000 * exp(-1.5*ln(78.3*ktemp)) * (sqrt(mue)/(1+
        sqrt(mue)) - 0.3*mue);
end;
```

```
function perH2CO3 (ph:real):real;
begin
```

W.3

```

perH2CO3:= 1/((1 + tento(-pk11)/tento(-ph) + tento(-pk11) * tento
(-pk22)/tento(-2*ph)));
end;

function perHCO3 (ph:real):real;
begin
perHCO3:= 1/(tento(-ph)/tento(-pk11) + 1 + tento(-pk22)/tento
(-ph));
end;

function perCO3 (ph:real):real;
begin
perCO3:=1/(tento(-2*ph)/(tento(-pk11)*tento(-pk22)) + tento
(-ph)/tento(-pk22) + 1);
end;

function per(ph,pkk:real):real;
begin
per:=tento(-pkk)/(tento(-pkk)+tento(-ph));
end;

function dH2CO3alk(pHf,pHs: real): real;
begin
dH2CO3alk:= perHCO3(pHf)-perHCO3(pHs) + 2 * (perCO3(pHf)-perCO3(pHs));
end;

function dHAcalk(pHf,pHs: real): real;
begin
dHAcalk:= per(pHf,pkaa) - per(pHs,pkaa);
end;

function MH2O (vxfi,vxs,pHfi,pHs: real):real;
begin
MH2O:= (vsdil+vxfs)*tento(-pHs)/tento(logf1) - (vsdil+vxfi)*tento(-pHfi)/
tento(logf1) + (vsdil+vxfi)*tento((pHfi-14)) - (vsdil+vxfs)*
tento((pHs-14));
end;

function MNH3(pHf,pHs:real):real;
begin
MNH3:= Nt/(14000*dil) * vsdil * (per(pHf,pKnn) - per(pHs,pKnn));
end;

function MHP04(pHf,pHs:real):real;
begin
MHP04:= Pt/(31000*dil) * vsdil * (per(pHf,pKpp) - per(pHs,pKpp))
end;

procedure mark;
begin
textbackground(white);
textcolor(black);
end;

procedure unmark;
begin
textbackground(black);
textcolor(white);
end;

procedure box(x1,y1,x2,y2,a:integer);
var
l1:integer;
begin
a:= x2-x1-1;
FOR l1:=1 to a DO
(draw the horizontal lines)

```

```

begin
  GOTOXY(x1+1,y1);
  writeln(chr(196));
  GOTOXY(x1+1,y2);
  writeln(chr(196));
  x1:=x1+1;
end;
x1:=x1-a;
a:=y2-y1-1;
FOR I1:=1 to a DO      (draw the vertical lines)
  begin
    GOTOXY(x1,y1+1);
    writeln(chr(179));
    GOTOXY(x2,y1+1);
    writeln(chr(179));
    y1:=y1+1;
  end;
y1:=y1-a;
GOTOXY(x1,y1);          (draw the corners)
writeln(chr(218));
GOTOXY(x1,y2);
writeln(chr(192));
GOTOXY(x2,y1);
writeln(chr(191));
GOTOXY(x2,y2);
writeln(chr(217));
end; {procedure box}

procedure introscreen;
begin
  clrscr; mark;
  box(10,3,70,20,1); unmark;
  box(24,6,56,8,1);box(22,5,58,9,1);
  box(26,10,55,16,1);
  GOTOXY(25,7); highvideo;
  write('FIVE pH POINT TITRATION METHOD');normvideo;
  GOTOXY(28,11);
  write('FOR DETERMINATION OF:');
  GOTOXY(28,13);
  write('(1) SHORT-CHAIN FATTY ACIDS');
  GOTOXY(28,15);
  write('(2) H2CO3*ALKALINITY');
  GOTOXY(11,16);
  GOTOXY(12,18);
  write('Copyright: UCT');
  box(10,22,41,24,1);
  GOTOXY(12,23);
  write('Press any letter to continue');
  intro := readkey;
end; {procedure introscreen}

procedure initlabel;
begin
  labels[1]:='pHo (initial pH).....';
  labels[2]:='pH1 (after adding Vx1).....';
  labels[3]:='pH2 (after adding Vx2).....';
  labels[4]:='pH3 (after adding Vx3).....';
  labels[5]:='pH4 (after adding Vx4).....';
  labels[6]:=' Vx1 (ml).....';
  labels[7]:=' Vx2 (ml).....';
  labels[8]:=' Vx3 (ml).....';
  labels[9]:=' Vx4 (ml).....';
  labels[10]:='Normality of titrant (mol/l)....';
  labels[11]:='Sample size: undiluted (ml).....';
  labels[12]:='Sample size: diluted (ml).....';
  labels[13]:='Temperature (Celsius).....';
  labels[14]:='TDS (mg/l).....';
  labels[15]:='Specific Conductivity (mS/m)....';
  labels[16]:='Inorganic Nitrogen (mgN/l).....';

```

```

    labels[17] := 'Inorganic Phosphorus (mgP/L)....';
end; {procedure initlabel}

procedure default_values;
begin
    value[1] := 7.36;
    value[2] := 6.75;
    value[3] := 5.95;
    value[4] := 5.18;
    value[5] := 4.29;
    value[6] := 1.06;
    value[7] := 3.50;
    value[8] := 4.84;
    value[9] := 5.40;
    value[10] := 0.0728;
    value[11] := 10;
    value[12] := 50;
    value[13] := 21;
    value[14] := 3300;
    value[15] := 488;
    value[16] := 0;
    value[17] := 0;
end; {procedure default_values}

procedure allocate_data;
begin
    pH0 := value[1];
    pH1 := value[2];
    pH2 := value[3];
    pH3 := value[4];
    pH4 := value[5];
    vx1 := value[6];
    vx2 := value[7];
    vx3 := value[8];
    vx4 := value[9];
    ca := value[10];
    vsundil := value[11];
    vsdil := value[12];
    temp := value[13];
    TDS := value[14];
    ionstr := value[15];
    Nt := value[16];
    Pt := value[17];
end; {allocate_data}

procedure write_value;
begin
    GOTOXY(37,2+posy);
    case posy of
        1,2,3,4,5,6,7,8,9: begin
            writeln(value[posy]:1:2);
        end;
        10: begin
            writeln(value[posy]:1:4);
        end;
        11,12,13,14,15,16,17,18: begin
            writeln(value[posy]:1:0);
        end;
        else
            end;
    end; {procedure write_value}

procedure screen;
begin
    mark; box(50,2,79,8,1); unmark;
    box(50,10,79,19,1);
    box(50,10,59,19,1);
    box(50,10,79,13,1);
    GOTOXY(52,4);

```

```

write(' 5 pH POINT TITRATION ');
GOTOXY(52,6);
write(' TITRATION INPUT DATA');
GOTOXY(52,12);
writeln(' KEY      FUNCTION');
GOTOXY(52,14);
writeln(' ',CHR(24),CHR(25),'  SELECT PARAMETER');
GOTOXY(52,15);
writeln('<enter>  ERASE VALUE');
GOTOXY(52,16);
writeln('<enter>  INSERT NEW VALUE');
GOTOXY(52,17);
writeln(' C      CALCULATION');
GOTOXY(52,18);
writeln(' Q      QUIT');
end; {procedure sreen}

```

```

procedure restore;
begin
  clrscr;
  screen;
  mark; box(1,1,47,22,1); unmark;
  FOR I:= 1 to 17 Do
    begin
      GOTOXY(3,2+I);
      writeln(labels[I]);
      GOTOXY(37,2+I);
      case I of
        1,2,3,4,5,6,7,8,9: begin
          writeln(value[I]:1:2);
          end;
        10: begin
          writeln(value[I]:1:4);
          end;
        11,12,13,14,15,16,17,18: begin
          writeln(value[I]:1:0);
          end;
      else
        end;
    end; {FOR}
  GOTOXY(3,2+posy);
  mark;
  write(labels[posy]);
  write_value;
  unmark;
end;

```

```

procedure display;
begin
  clrscr;
  screen;
  mark; box(1,1,47,21,1); unmark;
  FOR I:= 1 to 17 Do
    begin
      if I=1 then mark;
      GOTOXY(3,2+I);
      writeln(labels[I]);
      GOTOXY(37,2+I);
      case I of
        1,2,3,4,5,6,7,8,9: begin
          writeln(value[I]:1:2);
          end;
        10: begin
          writeln(value[I]:1:4);
          end;
        11,12,13,14,15,16,17,18: begin
          writeln(value[I]:1:0);
          end;
      else

```

```

end;
if I=1 then unmark;
end;
posy:=1;
REPEAT
repeat move := readkey until (upcase(move) in ['C','Q']) or (ORD(move)
in [72,80,13]);
case ORD(move) of
80: begin
GOTOXY(3,2+posy);
writeln(labels[posy]);
write_value;
posy:=posy+1;
if posy=18 then posy:=1;
GOTOXY(3,2+posy);
mark; writeln(labels[posy]); write_value; unmark;
end;
72: begin
GOTOXY(3,2+posy);
writeln(labels[posy]);
write_value;
posy:=posy-1;
if posy=0 then posy:=17;
GOTOXY(3,2+posy);
mark; writeln(labels[posy]); write_value; unmark;
end;
13: begin
repeat
GOTOXY(37,2+posy);
mark;
writeln(' ');
GOTOXY(37,2+posy);
{$I-} read(number) {$I+}
until ioresult = 0;
value[posy]:= number;
GOTOXY(37,2+posy);
unmark;
writeln(' ');
write_value;
case posy of
14: begin
value[posy+1]:= 0.1488*(value[posy]-20);
if value[posy+1] < 0 then value[posy+1]:= 0;
GOTOXY(37,posy+3);
write(' ');
GOTOXY(37,posy+3);
write(value[posy+1]:1:0);
end;
15: begin
value[posy-1]:= 6.72*value[posy]+20;
GOTOXY(37,posy+1);
write(' ');
GOTOXY(37,posy+1);
write(value[posy-1]:1:0);
end;
else
end;
end;
else
end;

until upcase(move) in ['C','Q'];
clrscr; writeln('Calculating, please wait');
end; {procedure display}

procedure pk ;
begin
ktemp:= 273 + temp;

```

```

if TDS < 20 then TDS:= 21;
logf1:= logf(mue(TDS,dil),ktemp);
logf2:= 4*logf1;
pK1:= -1*(-356.3094 - 0.06091964*ktemp + 21834.37/ktemp + 126.8339
*log(ktemp) - 1684915/(ktemp*ktemp));
pK11:= pK1 + logf1;
pK2:= -1*(-107.8871 - 0.03252849*ktemp + 5151.79/ktemp + 38.92561
*log(ktemp) - 563713.9/(ktemp*ktemp));
pK22:= pK2 - logf1 + logf2 ;
pKa:= 1170.5/ktemp - 3.165 + 0.0134 * ktemp;
pKaa:= pKa + logf1;
pKn:= 2835.8/ktemp -0.6322 + 0.00123 * ktemp;
pKnn:= pKn + logf1 ;
pKp:= 1979.5/ktemp - 5.3541 + 0.01984 * ktemp;
pKpp:= pKp - logf1 + logf2 ;
end; {procedure pK}

```

```

procedure deltaph;
begin
  pH0:= pH + pHcorr;  pH1:= pH1 + pHcorr;  pH2:= pH2 + pHcorr;
  pH3:= pH3 + pHcorr;  pH4:= pH4 + pHcorr;
end;

```

```

procedure atctcalculation;
begin
  A1:= (vx2-vx1)*ca - MH2O(vx1,vx2,pH1,pH2) - MNH3(pH1,pH2) -
  MHPO4(pH1,pH2) + dHAcalk(pH1,pH2)/dHAcalk(pH3,pH4) *
  (MH2O(vx3,vx4,pH3,pH4) + MNH3(pH3,pH4) + MHPO4(pH3,pH4) -
  (vx4-vx3)*ca);
  B1:= dH2CO3alk(pH1,pH2) - dHAcalk(pH1,pH2)/dHAcalk(pH3,pH4) *
  dH2CO3alk(pH3,pH4);

  A2:= (vx4-vx1)*ca - MH2O(vx1,vx4,pH1,pH4) - MNH3(pH1,pH4) -
  MHPO4(pH1,pH4) + dHAcalk(pH1,pH4)/dHAcalk(pH3,pH4) *
  (MH2O(vx3,vx4,pH3,pH4) + MNH3(pH3,pH4) + MHPO4(pH3,pH4) -
  (vx4-vx3)*ca);
  B2:= dH2CO3alk(pH1,pH4) - dHAcalk(pH1,pH4)/dHAcalk(pH3,pH4) *
  dH2CO3alk(pH3,pH4);

  Ct1:= (A1/B1)/vsdil * 50000 * dil;          {using pH: 1-2; 3-4}
  Ct2:= (A2/B2)/vsdil * 50000 * dil;          {using pH: 1-4; 3-4}
  MCt1:= A1/B1;
  Ctcomp:= Ct1 - Ct2 ;
end; {procedure atctcalculation}

```

```

procedure atct1;
begin
  dil:= vsdil/vsundil;
  pK; delpH:= 0; pHcorr:=0; atctcalculation; counter:= 0;
  MAT1:= 1/dHAcalk(pH3,pH4) * ((vx4-vx3)*ca-MCt1*dH2CO3alk(pH3,pH4) -
  MNH3(pH3,pH4) - MHPO4(pH3,pH4) - MH2O(vx3,vx4,pH3,pH4));
  At1:= MAT1/vsdil * 60000 * dil;
  CtAtratio:= At1/Ct1;
  if Ctcomp = 0 then x:= 'd';
  if Ctcomp > 0 then x:= 'a';
  if Ctcomp < 0 then x:= 'b';
  if CtAtratio > 0.5 then x:= 'c';
  case x of
    'a': begin
      pHcorr:= -0.01;
      repeat
        delpH:= delpH - 0.01;
        deltaph; counter:= counter + 1;
        atctcalculation;
      until (Ctcomp < 0) or (counter > 19);
      end;
    'b': begin
      pHcorr:= 0.01;
      repeat

```



```

    delpH:= delpH + 0.01;
    deltapH; counter:= counter + 1;
    atctcalculation;
    until (Ctcomp > 0) or (counter > 19);
end;
else
end;

Mat1:= 1/dHAcalk(pH3,pH4) * ((vx4-vx3)*ca-MCt1*dH2CO3alk(pH3,pH4) -
    MNH3(pH3,pH4) - MHPO4(pH3,pH4) - MH2O(vx3,vx4,pH3,pH4));
At1:= Mat1/vsdil * 60000 * dil;

Mat2:= 1/dHAcalk(pH1,pH4) * ((vx4-vx1)*ca-MCt1*dH2CO3alk(pH1,pH4) -
    MNH3(pH1,pH4) - MHPO4(pH1,pH4) - MH2O(vx1,vx4,pH1,pH4));
At2:= Mat2/vsdil * 60000 * dil;

MH2CO3alksam:= MCt1 * (perHCO3(pH0) + 2 * perCO3(pH0));
H2CO3alksam:= MH2CO3alksam / vsdil * dil * 50000 + (tento((pH0-14)) -
    tento(-pH0)/tento(logf1)) * 50000;
end; {procedure atct1}

procedure output;
begin
    clrscr;
    mark; box(10,2,60,4,1);unmark;
    box(2,7,73,10,1); box(49,7,73,10,1);
    box(2,11,73,14,1); box(49,11,73,14,1);
    box(2,15,73,18,1); box(49,15,73,18,1);
    GOTOXY(26,3);
    highvideo; write('OUTPUT DATA '); normvideo;
    GOTOXY(5,8);
    write('H2CO3*alkalinity (undiluted sample)');
    GOTOXY(58,8);
    highvideo; write(H2CO3alksam:3:0); normvideo;
    GOTOXY(5,9);
    write('(mg/l as CaCO3)');
    GOTOXY(5,12);
    write('Short-chain fatty acids (undiluted sample)');
    GOTOXY(58,12);
    if At1 > 0 then
        begin
            highvideo; write(At1:3:0); normvideo;
        end
    else
        begin
            highvideo; write('0 ');normvideo;
        end;
    GOTOXY(5,13);
    write('(mg/l as acetic acid)');
    GOTOXY(5,17);
    write('Systematic pH error');
    GOTOXY(58,17);
    highvideo; write(delpH:1:2); normvideo;
    if counter = 20 then
        begin
            GOTOXY(5,16);
            write('The titration data indicate a systematic');
            GOTOXY(5,17);
            write('pH error > 0.2; Check pH probe calibration');
            GOTOXY(58,17);
            write(' ');
        end;
    if x = 'c' then
        begin
            GOTOXY(5,16);
            write('Correction for systematic pH error');
            GOTOXY(5,17);
            write('is not possible for this titration');
            GOTOXY(60,17);
        end;
    end;
end;

```

```

    write(' ');
  end;
end; {procedure output}

```

```

procedure exitbox;
begin
  repeat;
    box(5,20,60,24,1); box(49,20,60,22,1);
    box(5,22,60,24,1); box(49,22,60,24,1);
    GOTOXY(7,21);
    write('Do you wish to do a further calculation ?');
    GOTOXY(53,21);
    highvideo; write('Y'); normvideo; write('/'); highvideo; write('N');
    normvideo;
    GOTOXY(7,23);
    write('Do you wish to quit the program ?');
    GOTOXY(54,23);
    highvideo; write('Q'); normvideo;
    again:=readkey; again:=upcase(again);
  until again in ['Y','N','Q'];
  clrscr;
end; {procedure exitbox}

```

```

begin
    ( of main program)

    introscreen;
    again:= 'Y';
    default_values;

    repeat; clrscr; initlabel; display;
    if upcase(move) in ['C'] then
      begin
        allocate_data; atct1; output; exitbox;
      end
    else
      begin
        clrscr; again:= 'N';
      end;
    until again in ['N','Q'];

end. (of main program)

```

(PROGRAM END.)